

Project Completion Report (PCR)

Project Name	Plutus Debugger VSCode Plugin
Project Number	1400098
Project Start Date	Nov 24, 2025
Project Completion Date	March 31, 2026

1. Context & Original Approach

Cardano Plutus development traditionally requires developers to rely on trace statements, compilation cycles, and raw node logs when diagnosing smart-contract behaviour. The project was undertaken in Fund 14 to reduce this friction by bringing interactive debugging and gas profiling directly into Visual Studio Code.

The approach was to build an open-source VSCode extension using the Debug Adapter Protocol (DAP), with practical developer tooling for breakpoints, real-time variable inspection, debugging, simulation, and gas/execution-unit profiling. The project also included documentation, security validation, tutorials, and a community launch workshop.

2. Deliverables & Outputs

We successfully delivered the core Plutus debugging and profiling extension together with its supporting documentation, testing, security, tutorial, and community-launch outputs. The project is open-source and released under the MIT License.

2.1 Core VSCode Extension (Open Source)

- **VSCode Extension:** A fully functional interactive debugging and gas-profiling extension for Cardano Plutus Smart Contracts.
- **Source Code:** Publicly hosted on GitHub as Kwired/Plutus-Debugger-VSCode-Plugin.
- **Distribution:** Published through the VSCode Marketplace and Open VSX Registry.
- **License:** MIT License.

2.2 Interactive Debugging & Profiling

- Developers can attach the debugger and set breakpoints during script execution.
- Variables can be evaluated and inspected in real time.
- Simulated execution traces can be used for gas/execution-unit profiling.
- The implementation includes unit testing of the Debug Adapter Protocol (DAP) functionality.

2.3 Ecosystem Features & Developer Workflow

- The delivered tooling supports debugging workflows intended for Cardano smart-contract developers, DApp auditors, and educators.
- The project evidence describes transaction simulation, testnet deployment support, and wallet emulation as part of the delivered product capability.
- The extension is designed to reduce repeated compile-and-log inspection cycles and make debugging substantially faster.

2.4 Documentation, Tutorials & Community Launch

- A public documentation portal was delivered for the plugin and its ecosystem outputs.
- Interactive tutorial material was provided through a YouTube playlist and Jupyter notebooks.
- A live ecosystem launch workshop was conducted with 118 developers.
- The workshop received 31 survey responses with an average rating of 4.1/5.0.

2.5 Security Validation

The extension underwent a security audit. The final evidence reports a clean result with zero critical vulnerabilities identified in the extension architecture.

3. Usage & Impact

The project targets Cardano smart-contract developers, DApp auditors, and educators. The available project completion evidence reports more than 500 downloads on the Open VSX Registry and early adoption by ecosystem developer teams.

Developer efficiency: Before the plugin, debugging commonly involved inserting trace statements, recompiling, and examining raw node logs. The delivered workflow reduces these debugging cycles from hours to minutes.

Community validation: The launch workshop provided direct beta-testing and community feedback. Participants reported significant reductions in debugging time, with the visual execution-unit profiler specifically highlighted as highly useful.

Ecosystem benefit: By providing professional-grade local debugging and profiling tools, the project aims to help developers produce higher-quality and more gas-efficient Plutus contracts.

4. Sustainability

The project is intended to remain an ongoing open-source tool. The codebase, releases, and documentation are permanently hosted through GitHub and GitHub Pages. Maintenance is planned through community pull requests and GitHub issue-based bug fixing.

The stated future roadmap includes Aiken trace support, a standalone CLI debugger, and cloud-based collaborative simulation environments.

5. Evidence & Links

Item	Evidence
GitHub Repository	https://github.com/Kwired/Plutus-Debugger-VSCode-Plugin
VSCode Marketplace	Plutus Debugger Extension
Open VSX Registry	kwired.plutus-debugger
Documentation Portal	Plutus Debugger Docs
Project Completion Video	https://youtu.be/QT1-W56BGxA
Interactive Tutorials — YouTube	https://www.youtube.com/playlist?list=PLRgDZ-VTHOrY
Interactive Tutorials — Jupyter	https://github.com/Kwired/Plutus-Debugger-VSCode-Plugin/tree/main/tutorials
Security Audit	Public final security audit report
Workshop Recording	https://youtube.com/live/DSBEHYquJas?feature=share
Workshop Evidence	Analytics and participant survey/feedback documents

6. Completion Summary

The Fund 14 project delivered an open-source VSCode extension focused on interactive Plutus debugging and gas profiling, supported by documentation, tutorials, security validation, and community engagement. The completion evidence records more than 500 Open VSX downloads, a live launch workshop with 118 developers and a 4.1/5.0 average survey rating, and a security audit with zero critical vulnerabilities.

This revised PCR follows the same high-level structure and presentation logic as the supplied Project Completion Report for Project 1400097, while using the facts and evidence specific to Project 1400098.